Separable Differential Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solving by separation of variables**

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

$y\,dy = x\,dx$, so $\frac{y^2}{2} = \frac{x^2}{2} + C$. Using $y(0) = 3$: $C = \frac{9}{2}$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (taking the positive root since $y(0) = 3 > 0$).

2 **[Foundational]** • **Target: Solving by separation of variables**

Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.

$\frac{1}{y}dy = 2x\,dx$, so $\ln|y| = x^2 + C$. Using $y(0) = 5$: $C = \ln 5$. Solution: $y = 5e^{x^2}$.

3 **[Foundational]** • **Target: Growth, decay, and applications**

A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.

a Solve for $P(t)$. $P(t) = 2000e^{0.05t}$.

b Find $P(20)$, to the nearest whole number. $P(20) = 2000e^{1} \approx 5437$.

(a) $P(t) = 2000e^{0.05t}$.; (b) $P(20) = 2000e^{1} \approx 5437$.

4 **[Foundational]** • **Target: Growth, decay, and applications**

Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a $20°$C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.

Solving: $T(t) = 20 + 80e^{-kt}$. Using $T(5) = 60$: $40 = 80e^{-5k}$, so $e^{-5k} = \frac{1}{2}$, giving $k = \frac{\ln 2}{5} \approx 0.1386$. So $T(t) = 20 + 80e^{-0.1386t}$.

5 **[Foundational]** • **Target: More separable equations**

Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.

$\frac{1}{y}dy = \frac{1}{x}dx$: $\ln|y| = \ln|x| + C$, so $y = Ax$ for constant A. Using $y(1) = 4$: $A = 4$: $y = 4x$.

6 **[Foundational]** • **Target: More separable equations**

Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

$\int\left(\frac{1}{y} + \frac{1}{1-y}\right)dy = \int dx$: $\ln|y| - \ln|1 - y| = x + C$, i.e. $\ln\left|\frac{y}{1-y}\right| = x + C$. Using $y(0) = \frac{1}{2}$: $\frac{y}{1-y} = 1$ at $x = 0$, so $C = 0$: $\frac{y}{1-y} = e^x$, giving $y = \frac{e^x}{1+e^x}$.

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7 [Foundational] • Target: Mixing and applied models

A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t. The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.

a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$). $6 = \frac{3y}{100}$ gives $y = 200$ kg.

b Explain, in words, what this equilibrium represents physically. In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

(a) $6 = \frac{3y}{100}$ gives $y = 200$ kg.; (b) In the long run, the salt content approaches 200 kg, at which point the rate of salt entering equals the rate leaving, keeping the amount constant.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Slope fields and qualitative behavior

For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.

At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.

9 [Intermediate] • Target: Slope fields and qualitative behavior

Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.

$\frac{dy}{dx} = 1 - 2e^{-x}$. Right side: $x - y = x - (x - 1 + 2e^{-x}) = 1 - 2e^{-x}$. Both sides equal $1 - 2e^{-x}$, confirming the solution.

10 [Intermediate] • Target: Visualizing a logistic solution curve

The graph shows the logistic solution $y = \frac{e^x}{1 + e^x}$ found earlier for $\frac{dy}{dx} = y(1 - y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \to \infty$ and as $x \to -\infty$.

As $x \to \infty$, $y \to 1$ (the curve levels off approaching the horizontal asymptote $y = 1$). As $x \to -\infty$, $y \to 0$. This matches the equilibrium solutions of $y(1 - y) = 0$ at $y = 0$ and $y = 1$.

11 [Intermediate] • Target: Another applied separable equation

Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.

$A(t) = 80e^{-kt}$. Using $A(30) = 20$: $\frac{20}{80} = \frac{1}{4} = e^{-30k}$, so $k = \frac{\ln 4}{30} \approx 0.0462$/day. Half-life: $e^{-kt_{1/2}} = \frac{1}{2}$ gives $t_{1/2} = \frac{\ln 2}{k} = \frac{30 \ln 2}{\ln 4} = 15$ days.

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12 [Intermediate] • Target: Separable equations with algebraic solutions

Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$, given $y(0) = 2$.

$y^2\,dy = x^2\,dx$: $\frac{y^3}{3} = \frac{x^3}{3} + C$. Using $y(0) = 2$: $\frac{8}{3} = C$. Solution: $y^3 = x^3 + 8$, i.e. $y = (x^3 + 8)^{1/3}$.

13 [Intermediate] • Target: Separable equations with algebraic solutions

Solve $\frac{dy}{dx} = \frac{1}{2y}$, given $y(0) = 3$.

$2y\,dy = dx$: $y^2 = x + C$. Using $y(0) = 3$: $C = 9$. Solution: $y = \sqrt{x + 9}$ (positive root, since $y(0) = 3 > 0$).

14 [Intermediate] • Target: Verifying a solution to an applied model

For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \to \infty$.

$\frac{dy}{dt} = 200(0.03)e^{-0.03t} = 6e^{-0.03t}$. Right side:
$6 - \frac{3}{100} \cdot 200(1 - e^{-0.03t}) = 6 - 6(1 - e^{-0.03t}) = 6e^{-0.03t}$. Both sides match. As $t \to \infty$, $e^{-0.03t} \to 0$, so $y(t) \to 200$, confirming the equilibrium.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.

$e^y\,dy = e^x\,dx$: $e^y = e^x + C$. Using $y(0) = 0$: $1 = 1 + C$, so $C = 0$: $e^y = e^x$, giving $y = x$.

16 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P\left(1 - \frac{P}{500}\right)$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.

Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 500$. Starting from $P(0) = 50 > 0$, the population will grow toward the stable equilibrium $P = 500$ (the carrying capacity), since $P = 0$ is an unstable equilibrium for any positive starting population.

17 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 6.01 Related Rates

Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.

$y\,dy = \cos x\,dx$: $\frac{y^2}{2} = \sin x + C$. Using $y(0) = -2$: $2 = 0 + C$, so $C = 2$: $y^2 = 2\sin x + 4$, giving $y = -\sqrt{2\sin x + 4}$. The negative root is required because $y(0) = -2 < 0$, and y must vary continuously from that starting sign.

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18 [Challenging] • Target: A final applied separable model • Cross-Lesson: 6.02 Optimization and Curve Sketching

A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.

$C(t) = 50e^{-0.2t}$. Setting $50e^{-0.2t} = 5$: $e^{-0.2t} = 0.1$, so $t = \frac{\ln 0.1}{-0.2} = \frac{-2.3026}{-0.2} \approx 11.5$ hours.

19 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 7.01 Integration by Substitution

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:.

Final Answer: maximum. Applying Differential Equations principles, Maximum growth at $P = K/2$. Therefore, the verified correct option is maximum.

20 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 7.02 Integration by Parts

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$.

Final Answer: 0. Applying Differential Equations principles, At $P = K$: $dP/dt = rK(1 - K/K) = 0$. Therefore, the verified correct option is 0.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Solutions of Separable Equations

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$.

a. $y^2 = 2x^2 + C$ ✓

b. $y = x^2 + C$

c. $y^2 = x^2 + C$

d. $y = 2x^2 + C$

Separate variables: $y\,dy = 2x\,dx$. Integrate both sides: $\frac{1}{2}y^2 = x^2 + C_1 \Rightarrow y^2 = 2x^2 + C$.

22 [Foundational] • Target: General Solutions of Separable Equations

Find the general solution to $\frac{dy}{dx} = 3x^2y$.

a. $y = Ce^{x^3}$ ✓

b. $y = e^{x^3} + C$

c. $y = Cx^3$

d. $y = 3x^3 + C$

$\frac{1}{y}\,dy = 3x^2\,dx \Rightarrow \ln|y| = x^3 + C_1 \Rightarrow |y| = e^{C_1}e^{x^3} \Rightarrow y = Ce^{x^3}$.

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23 [Foundational] • Target: General Solutions of Separable Equations

Find the general solution of $\frac{dy}{dx} = (1 + y^2)e^x$.

a. $\arctan(y) = e^x + C$ ✓

b. $y = \tan(e^x) + C$

c. $y = e^x + C$

d. $\ln(1 + y^2) = e^x + C$

$\frac{1}{1+y^2}\,dy = e^x\,dx \Rightarrow \arctan(y) = e^x + C$.

24 [Foundational] • Target: Initial Value Problems (IVPs)

Solve the initial value problem $\frac{dy}{dx} = 2xy$ with $y(0) = 3$.

a. $y = 3e^{x^2}$ ✓

b. $y = e^{x^2} + 2$

c. $y = 3e^{2x}$

d. $y = 3x^2 + 3$

$\frac{dy}{y} = 2x\,dx \Rightarrow \ln|y| = x^2 + C_1 \Rightarrow y = Ce^{x^2}$. Using $y(0) = 3 \Rightarrow C = 3$. Thus $y = 3e^{x^2}$.

25 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dx} = -\frac{x}{y}$ with $y(3) = 4$.

a. $x^2 + y^2 = 25$ ✓

b. $x^2 - y^2 = -7$

c. $y = -x + 7$

d. $x^2 + y^2 = 7$

$y\,dy = -x\,dx \Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C_1 \Rightarrow x^2 + y^2 = C$. At $(3, 4)$: $9 + 16 = 25 \Rightarrow x^2 + y^2 = 25$.

26 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dt} = ky$ given $y(0) = 50$ and $y(2) = 200$. What is the formula for $y(t)$?

a. $y(t) = 50 \cdot 2^t$ ✓

b. $y(t) = 50e^{2t}$

c. $y(t) = 50 \cdot 4^t$

d. $y(t) = 50 + 75t$

$y(t) = y_0e^{kt} = 50e^{kt}$. $y(2) = 50e^{2k} = 200 \Rightarrow e^{2k} = 4 \Rightarrow e^k = 2$. Thus $y(t) = 50(2^t)$.

27 [Foundational] • Target: Slope Fields and Solution Curves

Along which curve in the xy-plane do the line segments in the slope field for $\frac{dy}{dx} = x - y$ have slope 0?

a. **Along the line** $y = x$ ✓

b. Along the x-axis ($y = 0$)

c. Along the y-axis ($x = 0$)

d. Along $y = -x$

Slope $= 0 \Rightarrow \frac{dy}{dx} = 0 \Rightarrow x - y = 0 \Rightarrow y = x$.

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28 [Foundational] • Target: Slope Fields and Solution Curves

A slope field shows line segments that depend ONLY on y (horizontal rows of equal slopes). Which differential equation could this represent?

a. $\frac{dy}{dx} = y(2 - y)$ ✓

b. $\frac{dy}{dx} = x^2 - y$

c. $\frac{dy}{dx} = xy$

d. $\frac{dy}{dx} = 3x + 1$

If slopes are identical for every x at a fixed y, $\frac{dy}{dx}$ is independent of x, as in autonomous equations like $\frac{dy}{dx} = y(2 - y)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Slope Fields and Solution Curves

For $\frac{dy}{dx} = x^2 + y^2$, what can be said about all non-origin slopes in the slope field?

a. **All slopes are strictly positive** ✓

b. Slopes are positive in quadrants I and III only

c. Slopes are 0 everywhere

d. Slopes are negative when $x < 0$

Since $x^2 + y^2 > 0$ for all $(x, y) \neq (0, 0)$, all tangent line segments have strictly positive slope.

30 [Intermediate] • Target: Logistic Growth Models

In the logistic differential equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$, what is the carrying capacity M?

a. 800 ✓

b. 0.05

c. 400

d. 16000

The standard logistic equation is $\frac{dP}{dt} = kP(1 - P/M)$, where M is the carrying capacity. Here $M = 800$.

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31 [Intermediate] • Target: Logistic Growth Models

At what population P is the growth rate $\frac{dP}{dt}$ maximized in the logistic equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$?

a. $P = 400$ ✓

b. $P = 800$

c. $P = 200$

d. $P = 0$

The growth rate is a quadratic function of P that attains its maximum at half the carrying capacity: $P = M/2 = 800/2 = 400$.

32 [Intermediate] • Target: Logistic Growth Models

What is $\lim_{t \to \infty} P(t)$ for any solution to $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$ with $P(0) = 50$?

a. 800 ✓

b. ∞

c. 400

d. 50

For any initial population $P_0 > 0$, the logistic curve asymptotically approaches the stable carrying capacity $M = 800$ as $t \to \infty$.

33 [Intermediate] • Target: Slope Fields and Solution Curves

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$ ✓

Applying Differential Equations principles, $k > 0$: growth; $k0$.

34 [Intermediate] • Target: Logistic Growth Models

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$ ✓

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

The given differential equation is $\frac{dy}{dx} = 2x$. To find the general solution, we integrate both sides with respect to x: $\int \frac{dy}{dx} dx = \int 2x dx$ $y = x^2 + C$, where C is the constant of integration. We are given that the particular solution passes through the point $(1, 3)$. Substitute these values into the general solution: $3 = (1)^2 + C$ $3 = 1 + C$ $C = 3 - 1 = 2$. Therefore, the particular solution is $y = x^2 + 2$.

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35 [Intermediate] • Target: Solving by separation of variables

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1 e^{r_1 x} + C_2 e^{-r_1 x}$

b. $y = C_1 e^{r_1 x}$

c. $y = (C_1 + C_2)e^{2r_1 x}$

d. $y = (C_1 + C_2 x)e^{r_1 x}$ ✓

Applying Differential Equations principles, Repeated root: multiply by x. Therefore, the verified correct option is $y = (C_1 + C_2 x)e^{r_1 x}$.

36 [Intermediate] • Target: Growth, decay, and applications

$y'' + y = 0$: general solution:

a. $y = C_1 \sin x + C_2 \cos x$ ✓

b. $y = C_1 e^{x} + C_2 e^{-x}$

c. $y = C_1 x + C_2$

d. $y = C_1 e^{-x}$

Applying Differential Equations principles, Complex roots $\pm i$. Therefore, the verified correct option is $y = C_1 \sin x + C_2 \cos x$.

37 [Intermediate] • Target: More separable equations

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$ ✓

d. $-\cos x$

Applying Differential Equations principles, $\int \tan x \, dx = -\ln|\cos x|$. Therefore, the verified correct option is $-\ln|\cos x|$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Mixing and applied models • Cross-Lesson: 6.02 Optimization and Curve Sketching

First-order linear ODE solution structure: general solution = homogeneous solution +

a. **particular solution** ✓

b. another homogeneous solution

c. the initial condition

d. zero

Applying Differential Equations principles, $y = y_h + y_p$. Therefore, the verified correct option is particular solution.

39 [Challenging] • Target: Slope fields and qualitative behavior • Cross-Lesson: 7.01 Integration by Substitution

Logistic model: $dP/dt = 0.1P(1 - P/100)$, $P(0) = 10$. $P \to$ as $t \to \infty$:

a. 0

b. 10

c. ∞

d. 100 ✓

Applying Differential Equations principles, Carrying capacity $K = 100$. Therefore, the verified correct option is 100.

40 [Challenging] • Target: Visualizing a logistic solution curve • Cross-Lesson: 7.02 Integration by Parts

General solution of $\frac{dy}{dx} = 2x$:

a. $y = x^2$

b. $y = 2$

c. $y = x^2 + C$ ✓

d. $y = 2x + C$

Applying Differential Equations principles, Integrate: $y = x^2 + C$. Therefore, the verified correct option is $y = x^2 + C$.

41 [Challenging] • Target: Another applied separable equation • Cross-Lesson: 8.01 Area Between Curves

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$ ✓

c. $y = 2x + 2$

d. $y = xe^x$

Applying Differential Equations principles, $y = Ce^x$; $y(0) = C = 2$. Therefore, the verified correct option is $y = 2e^x$.

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42 [Challenging] • Target: Separable equations with algebraic solutions • Cross-Lesson: 8.02 Volumes of Revolution

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$ ✓

Applying Differential Equations principles, $N = N_0 e^{-kt}$. Therefore, the verified correct option is $N = N_0 e^{-kt}$.

43 [Challenging] • Target: Verifying a solution to an applied model • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Separable ODE $\frac{dy}{dx} = \frac{y \ln y}{x}$, solution:

a. $\ln y = Cx$

b. $y = Ae^{Bx}$

c. $y = e^{Cx}$

d. $\ln(\ln y) = \ln x + C \Rightarrow \ln y = Ax$ ✓

$\int \frac{dy}{y \ln y} = \int \frac{dx}{x} \Rightarrow \ln(\ln y) = \ln x + C \Rightarrow \ln y = Ax.$

44 [Challenging] • Target: A final applied separable model • Cross-Lesson: 1.01 Precalculus Review for Calculus

Separable ODE: $\frac{dy}{dx} = \frac{y}{x}$. Solution:

a. $y = x + C$

b. $y = Ce^x$

c. $y = Cx$ ✓

d. $y = C/x$

$\frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln|y| = \ln|x| + C \Rightarrow y = Cx.$

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45 [Challenging] • Target: General Solutions of Separable Equations • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$ ✓

Euler: $e^{\alpha x}$ envelope times sinusoids.

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